

Credit - Drishti Ias

Epigenetic Inheritance



Source: TH

Why in News?

A study published in *Nature Genetics* found that certain **epigenetic changes, particularly DNA methylation patterns, can be inherited across generations in mice without altering the underlying DNA sequence** .

- The findings challenge the conventional understanding of **Mendelian inheritance** and suggest that heritable information may extend beyond the DNA sequence.
- However, since the study was conducted in **mice** , the findings cannot be directly extrapolated to humans.

What are the Key Findings of the Study on Epigenetic Changes?

- **Non-Mendelian Inheritance:** Researchers identified around **7,600 genomic locations with different DNA methylation patterns** , of which **93% followed Mendelian inheritance** , while about **7% showed non-Mendelian epigenetic inheritance** , accounting for **more than 500 instances** .
 - The findings suggest that **epigenetic inheritance may be much more widespread than previously recognised** .
- **Sex-Specific Epigenetics:** In over 300 areas of the mouse genome, researchers observed that **methylation patterns were dependent on the animal's sex**, with females showing significantly **higher methylation in their liver tissues than males**.
- **Discovery of New Genes:** The study uncovered at least five new genes that display genomic imprinting.
- **Naturally Occurring Paramutation:** Researchers identified **naturally occurring intergenerational paramutation** (a process where one gene copies its epigenetic marks onto another), at the **Capn11 gene** , which is involved in **testicular meiosis and male fertility** .
 - They also found **two additional highly likely cases of paramutation** involving **intracisternal A particles (IAPs) transposable elements** .
 - **IAPs** are **endogenous retroviral elements embedded in the mouse genome** that can **resist normal DNA demethylation during sperm and egg formation** , potentially allowing epigenetic changes to persist across generations.

What is Epigenetics?

- **About:** Epigenetics is the study of **heritable changes in gene expression** (how genes are turned on or off) that occur **without any alteration to the underlying DNA sequence**.
- **Core Concept:** If the DNA sequence is the "hardware" or the text of an instruction manual, the epigenome is the "software" or the highlighter that determines which parts of the manual are actually read and executed by the cell.
 - Though cells contain the same DNA, epigenetic differences allow **liver, brain and muscle cells** to perform different functions.
 - Epigenetic markers can change with **diet, stress, ageing, physical activity and toxins**, linking environmental factors to biological outcomes.
- **Epigenetic mechanisms:**
 - **DNA Methylation:** This is the most common epigenetic mechanism. It involves the addition of a **chemical tag called a methyl group (-CH₃)** directly to the DNA, typically at cytosine bases.
 - DNA methylation, particularly at gene promoters, is often associated with reduced gene transcription.
 - **Histone Modification:** DNA is not loose in the nucleus; it is wrapped around protein spools called **histones to form chromatin**. Chemical modifications to these histones dictate how tightly the DNA is packed.
 - Tightly packed DNA is inaccessible and therefore silenced.
 - Loosely packed DNA is accessible for transcription, meaning the gene is active.
 - **Non-coding RNAs:** These are functional RNA molecules that are transcribed from DNA but do not translate into proteins.
 - They can bind to coding RNAs to break them down, preventing them from producing proteins and thereby regulating gene expression.
- **Genomic Imprinting:** It is an epigenetic phenomenon in which gene expression depends on whether a gene copy was inherited from the mother or the father.
 - Normally, mammals **inherit two working copies (alleles)** of most genes—one from the mother and one from the father.
 - In imprinted genes, one of these copies is **epigenetically silenced** (usually via methylation) before fertilization. For example, if a specific gene is maternally imprinted, only the paternal copy will be active in the offspring.
- **Paramutation:** It is a process in which the epigenetic state of one allele alters that of its homologous allele, with the altered state potentially persisting across generations..
- **Implications:**
 - **Addressing Missing Heritability:** Epigenetic changes may explain some heritable traits that traditional **Genome-Wide Association Studies (GWAS)** cannot detect because they do not involve changes in DNA sequence.
 - **Precision Medicine:** Reversible epigenetic changes could be targeted through **epigenetic therapies** to restore normal gene function.
 - **Environment & Inheritance:** Factors such as **diet, stress and toxins can influence epigenetic marks**, highlighting the importance of **nutrition, environmental protection and parental health**.
 - The extent to which such changes are inherited across generations in humans remains under study.

- **Complex Disease Pathology:** Non-Mendelian inheritance patterns provide a structural explanation for diseases that resist traditional genetic tracking, including **specific cancers, metabolic syndromes, and conditions that display strict sex-dependent inheritance**.
- **Evolution:** Epigenetic changes may enable faster adaptation to environmental changes than genetic mutations.

Epigenetic Inheritance Differ from Mendelian Inheritance

Feature	Conventional Mendelian Inheritance	Epigenetic Inheritance
Basis	Transmission of DNA sequence variants (alleles) from parents to offspring.	Transmission of gene-regulating epigenetic marks without changing DNA sequence.
Key Principle	Follows Mendel’s laws of segregation and independent assortment.	Can show non-Mendelian inheritance patterns .
Gene Expression	Primarily determined by the inherited DNA sequence .	Influenced by epigenetic marks that can switch genes on or off .
Stability	DNA sequence is generally stable across generations .	Epigenetic marks are dynamic and reversible .
Reproductive Resetting	DNA sequence is transmitted through sperm and eggs .	Most epigenetic marks are normally erased during formation of sperm and eggs .
Inheritance Pattern	Generally follows predictable parent-to-offspring patterns .	Some marks can escape resetting, vary by sex, or affect another allele through paramutation.

THEORIES OF EVOLUTION

The modification of living organisms during their descent, generation by generation from common ancestors.

Oparin-Haldane Theory of Origin of Life

- Also known as Materialistic theory
- Describes process of origin of life on early Earth as:

Physio-chemical processes of atoms → Organic compounds → Macromolecules → First living system or cells

Theory of Inheritance of Acquired Character (Lamarckism)

- First theory of organic evolution
- Evolutionary ideas:**
 - Internal forces of life increase the size of organism
 - New structures appear because of an 'inner want'
 - Direct environmental effect over living organisms
 - Inheritance of acquired character
- E.g.;** Long neck of giraffe due to gradual lack of surface vegetation

Theory of Natural Selection (Darwinism)

- Foundation of evolutionary biology
- Elements:**
 - Universal occurrence of variation
 - Rapid multiplication
 - The struggle for existence** - Intraspecific and interspecific
 - Survival of the fittest (Natural Selection)**
 - Inheritance of useful variations; Elimination of non-useful variations
- E.g.;** Survival of more dark-winged moths than white-winged ones in post-industrialisation period

Neo-Darwinism

Integration of Darwin's theory of evolution with Gregor Mendel's theory of genetics

Modern Synthetic Theory

- One of the proven theories of organic evolution
- Includes factors such as – Mutation, Variation / Recombination, Heredity, Natural Selection and Isolation

Mutation Theory (Hugo de Vries)

- Describes evolution as a jerky process where new varieties of species are formed by mutations (discontinuous variations)
- Salient features:**
 - Mutation appears all of a sudden and becomes operational immediately
 - Same type of mutation in several individuals of a species
 - All mutations are inheritable
 - Useful mutations are selected and lethal ones are eliminated by nature



Click here to Read: [Lamarckian Inheritance and Epigenetics Evolution](#)

Frequently Asked Questions

1. What is epigenetics?

Epigenetics refers to changes in **gene expression without altering the underlying DNA sequence**, mainly through mechanisms such as **DNA methylation and histone modification**.

2. What is genomic imprinting?

Genomic imprinting is an epigenetic phenomenon in which gene expression depends on whether an allele is inherited from the **mother or father**.

3. What is paramutation?

Paramutation occurs when the epigenetic state of one allele alters another allele and the altered state can **persist across generations**.

4. How can environmental factors influence epigenetics?

Factors such as **diet, stress, toxins and ageing** can modify epigenetic marks, potentially affecting **gene expression and health**.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/muHg-KDIJ7s>

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UPSC Civil Services Examination, Previous Year Questions (PYQs)

Q. What is Cas9 protein that is often mentioned in the news? (2019)

- (a) A molecular scissors used in targeted gene editing
- (b) A biosensor used in the accurate detection of pathogens in patients
- (c) A gene that makes plants pest-resistant
- (d) A herbicidal substance synthesized in genetically modified crops

Ans: (a)

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future? (2017)

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (d)

Mains

Q. What are the research and developmental achievements in applied biotechnology? How will these achievements help to uplift the poorer sections of society? **(2021)**

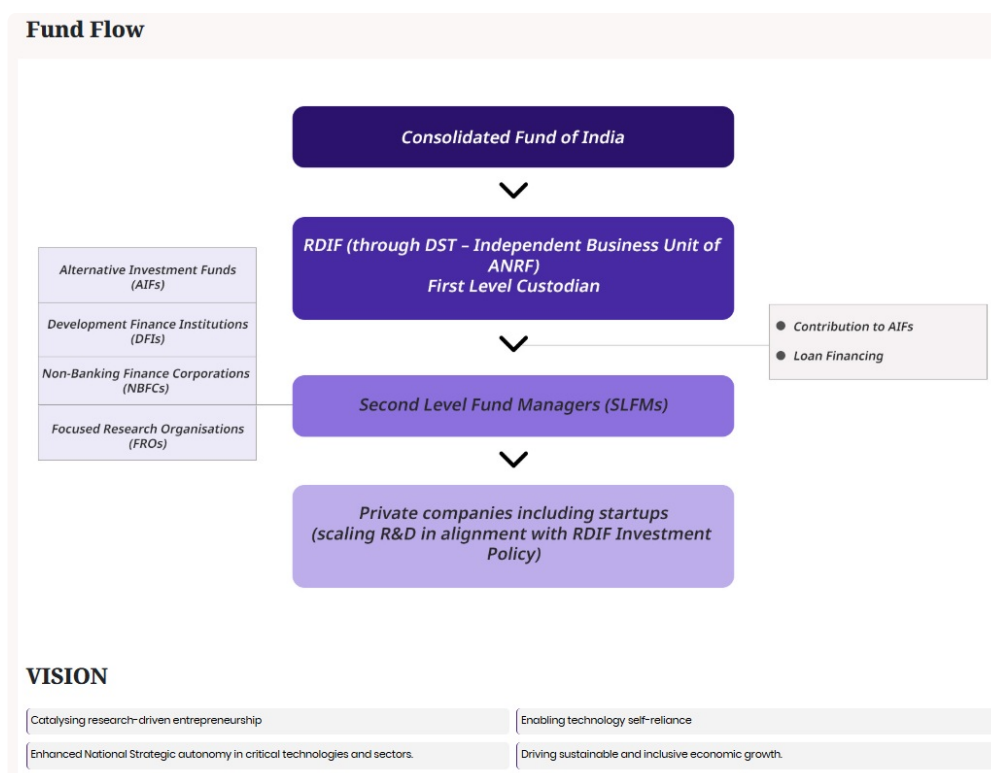
Research, Development and Innovation Fund



Source: IE

The government's Rs 1 lakh crore **Research, Development and Innovation Fund (RDIF)** has come into focus after concerns were raised over the selection process for companies receiving concessional loans under the flagship deep-tech financing initiative.

- **About:** The **RDIF** is a **flagship initiative** under the **Department of Science and Technology (DST)** and a **Special Purpose Fund** under the **Anusandhan National Research Foundation (ANRF)** .
 - Approved by the Union Cabinet in 2025, the RDI Scheme has a Rs 1 lakh crore outlay over six years to **catalyse private sector investment** in research and development (R&D).
- **Key Features:** The fund provides collateral-free, low-cost, long-term financing for high-risk, high-impact projects (Technology Readiness Level (TRL 4) and above) in **sunrise and strategic sectors** , including AI, quantum technology, semiconductors, space, defence, robotics, clean energy, and digital healthcare.
- **Objectives:** It aims to **promote research-driven entrepreneurship, strengthen technology self-reliance, support acquisition of critical technologies, establish a dedicated Deep-Tech Fund of Funds, enhance strategic autonomy, and drive sustainable economic growth** .
- **Funding Mechanism:** Funding is channelled through **Alternative Investment Funds (AIFs), Development Finance Institutions (DFIs), Non-Banking Financial Companies (NBFCs), and Focused Research Organisations (FROs)** , which invest in **Indian companies and startups developing RDI-intensive technologies** .
 - Loans under the **RDI Fund** are routed through designated **Second-Level Fund Managers (SLFMs)** , which also select the **Eligible Technology Entities (ETEs)** for funding.
 - Currently, the two SLFMs are **Technology Development Board (TDB)** under the DST and **Biotechnology Industry Research Assistance Council (BIRAC)** , under the Department of Biotechnology.



Read more: [Research, Development and Innovation \(RDI\) Scheme](#)

Supreme Court Directs Measures to Curb 'Digital Arrest' Scams



Source: TH

Recently, the **Supreme Court** directed the Centre, States/UTs, RBI, telecom authorities and investigating agencies to **adopt coordinated measures to curb “ digital arrest ” scams** and ensure timely recovery of victims' money.

- **SC Directions:** The Supreme Court directed the **RBI to issue an SOP within four weeks for temporary debit** holds on mule accounts linked to **cyber fraud or money laundering** and share it with the **Registrars General of High Courts** .
 - States and UTs were directed to establish **grievance-redressal and money-restoration mechanisms** for cyber-fraud victims in line with the **Ministry of Home Affairs' SOP for the National Cyber Crime Reporting Portal and Cyber RMS**.
- **Digital Arrest:** Digital arrest is a **cyber scam** in which fraudsters **impersonate law-enforcement agencies** to extort money by creating fear and urgency.
 - Scammers contact victims, falsely link them to criminal cases, and threaten arrest **or freezing of bank accounts** using spoofed numbers and fake documents.
 - They exploit this fear to pressure victims into paying a **fake fine or security deposit** to avoid the fabricated legal action.
- **Decline:** According to the **Indian Cyber Crime Coordination Centre (I4C)**, digital-arrest complaints declined sharply from 1,23,672 in 2024 to 58,249 in 2025 and 16,377 by June 2026.
- **Initiatives to Tackle Digital Arrest:**
 - **Money Restoration Mechanism:** The **Money Restoration Mechanism**, supported by a data-sharing MoU between **RBI Innovation Hub and I4C**, now covers 1,23,590 branches of 69 banks, with 57 banks participating nationwide and ₹18.05 crore restored in 36,290 cases.
 - **e-Zero FIR mechanism:** The **e-Zero FIR mechanism** is operational in 19 States, while only 14 States have notified Cyber Crime Coordination Centres, prompting the Supreme Court to **direct** all remaining States/UTs to implement both within four weeks.
 - **Telecom Rules:** The **Telecommunications (Radio Equipment Possession Authorisation) Rules, 2025** have been notified, while the Telecommunications (User Identification) Rules, 2025 are in the final stage of notification.
 - **Committee Recommendations:** The **Inter-Departmental Committee** asked the **CBI to consider lowering the ₹10-crore threshold and take up cases** where multiple frauds by the same organised network cumulatively exceed this limit.

Read More: [SC Directions on 'Digital-Arrest' Scams](#)

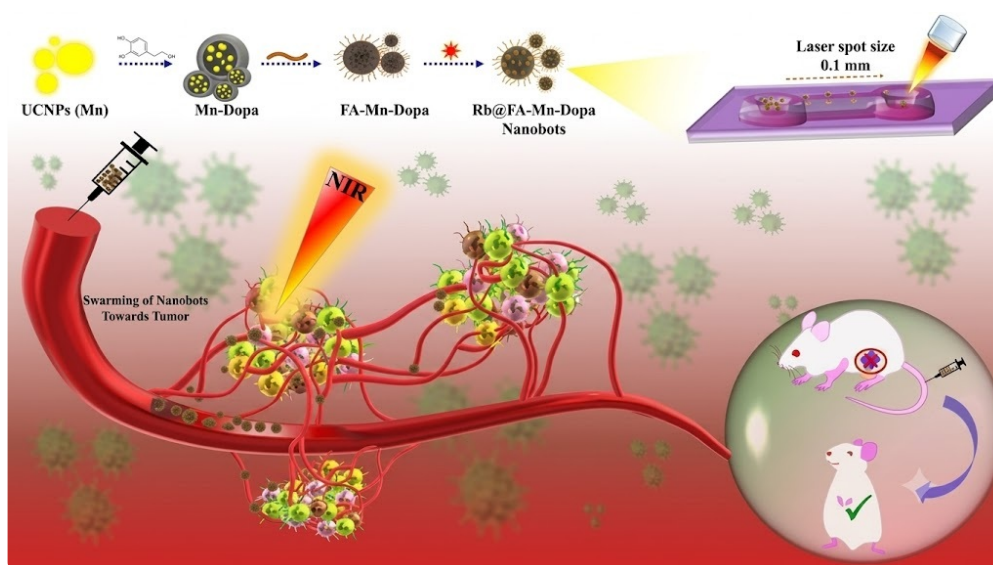
Light-Activated Nanobots for Targeted Breast Cancer Therapy



Source: PIB

Scientists from the **Institute of Nano Science and Technology (INST), Mohali** have developed **light-activated multifunctional nanobots** for more precise and targeted breast cancer therapy.

- **Working:** The **upconversion nanoparticle (UCNP)-based nanobots** respond to near **-infrared light** and exhibit phototaxis, enabling directional movement toward the light source for localized therapeutic action.
- **Cancer Cell Targeting:** **Folic acid** functionalisation enables the nanobots to selectively recognise **breast cancer** cells overexpressing folate receptors, while irradiation with a **980-nanometre near-infrared laser** produces localized heating to destroy cancer cells.
- **Photodynamic Therapy:** A photosensitizer generates **reactive oxygen species (ROS)** under near-infrared irradiation, with combined photothermal and photodynamic effects producing greater tumour inhibition than either therapy alone.
- **Smart Control:** The nanobots' therapeutic action can be influenced by **laser intensity, pH and glutathione concentration** , enabling better control in biological environments.
- **Key Advantage:** The fuel-free nanobots enable **externally controlled and localized cancer treatment** while potentially **reducing damage to healthy tissues and systemic toxicity** .
- **Testing & Significance:** The nanobots **showed therapeutic efficacy in cellular models and breast tumour-bearing mice** , highlighting their potential for precise, targeted and minimally invasive breast cancer therapy.



Read More: [Nanorobots in Healthcare](#)

